



Betsy Johanna Villa Brochero

PhD Candidate in Computer Vision – University of Twente

PhD candidate in Computer Vision at the University of Twente working on real time defect detection in industrial assembly processes using machine learning and visual inspection systems. Background in mechatronics, robotics, and industrial AI applications with experience deploying computer vision systems in manufacturing environments.

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EDUCATION

PhD in Computer Vision University of Twente

09/2023 - Present

Research focus:

- Computer vision for real-time defect detection in industrial assembly processes
- Improving reliability and quality assurance in manufacturing environments
- Machine learning models for automated visual inspection systems

Master of Science in Mechatronics and Micromechatronics Systems Oviedo University/ Karlsruhe University of Applied Sciences

09/2018 - 09/2020

Spain/Germany

Relevant Areas

- Robotics, Artificial Intelligence, Control Systems, Mechatronics.
- Master Thesis: Control and Monitoring of an induction hardening test bench.

Bachelor of Science in Mechatronics Engineering

Autonoma del Caribe, University

01/2010 - 03/2015

Colombia

Bachelor Thesis

- Bachelor Thesis : Digital image processing applied on static sign language recognition system.

LANGUAGES

English
Full Professional Proficiency

German
Professional Working Proficiency

Spanish
Native or Bilingual Proficiency

Dutch
Limited Working Proficiency

French
Limited Working Proficiency

TECHNICAL SKILLS

Programming

Python, C++, MATLAB, Arduino, Java

Computer Vision & ML

OpenCV, image processing, defect detection systems, industrial vision systems

Engineering Tools

LabVIEW, Simulink

RESEARCH INTERESTS

Computer Vision for Industrial Inspection

Deep Learning for Defect Detection

Real Time Vision Systems

Visual Quality Control in Manufacturing

Machine Learning for Industrial Automation

RESEARCH PROJECTS

Visual Inspection Assembly System (VIAS) – Gunnebo Safe Storage

- Designed and deployed a computer vision system for automated inspection in assembly processes.
- Implemented machine learning algorithms for defect and missing-object detection.
- Integrated real-time monitoring into manufacturing environments.
- Enabled data-driven quality analysis and improved inspection consistency.

Synaps – Digital Manufacturing Environment Platform

- Led integration of manufacturing systems into a centralized digital platform.
- Enabled real time production data monitoring and cross departmental visibility.
- Improved traceability and production analytics through system integration.

RESEARCH EXPERIENCE

PhD Research – Computer Vision for Industrial Inspection

University of Twente

09/2023 - Present

Tasks

- Developing deep learning and computer vision algorithms for automated defect detection in industrial assembly processes.
- Designing machine learning pipelines for visual inspection systems.
- Integrating real-time monitoring solutions into industrial production environments.
- Investigating robustness and scalability of inspection models in manufacturing settings.

Supervisor : Dr. Estefania Talavera - e.talaveramartinez@utwente.nl

PROFESSIONAL EXPERIENCE

Industrial Transformation Engineer (Global)

Gunnebo Safe Storage

09/2023 - Present

Tasks

- Driving Industry 4.0 initiatives across global manufacturing sites.
- Developing digitalization roadmaps for automation and data integration.
- Supporting deployment of AI-based inspection systems and production analytics tools.

Automation & Digital Transformation Engineer

Gunnebo Safe Storage

08/2022 - 08/2023

Tasks

- Implemented Industry 4.0 solutions, including IoT, data acquisition, and machine learning.
- Developed automation projects, improving production efficiency and quality monitoring.

Trainee Digital Transformation Engineering

Gunnebo Safe Storage

03/2021 - 07/2022

Netherlands

The Gunnebo Group is a global provider of security products, services and software

Tasks

- Designed automated visual inspection systems for assembly lines.
- Developed machine learning models for defect detection.

PUBLICATIONS

Paper

Digital image processing applied on static sign language recognition system

Author(s)

B.Villa et al.

2018

Prospectiva/Issue 16/ Pag. 41-48.

Paper

Computer Vision in Manufacturing for Missing Object Detection: A Survey

Work in Progress

Conference Paper

PCB-MC: A Benchmark Dataset for Missing Component Detection in Printed Circuit Boards

Work in Progress

AWARDS & SCHOLARSHIPS

Erasmus Mundus Joint Master Scholarship Mechatronics (2018 - 2020)

European Union

- Full scholarship to pursue a joint master's degree in Mechatronics at Karlsruhe University of Applied Sciences and University of Oviedo.

Multidisciplinary Artificial Intelligence Training Program (2021 - 2022)

IALAB- Buenos Aires University, Argentina

- Training program in Artificial Intelligence funded by the Development Bank of Latin America.

DAAD Scholarship for study trips to Germany (2014)

DAAD

- Academic mission visiting research centers and universities in Germany.